

Algebraic Coincidence Classes

A, B, C, D, E in ECDSA

Identification Formulas, Private-Key Recovery
Formulas, and Worked Examples

This document gives a complete mathematical description of the five coincidence classes used by the ECDSA-Research classifier: the exact conditions that define each class, the closed-form private-key recovery formulas for classes A, B and E, and the structural reasons why classes C and D admit none. Every formula is exercised on the signatures shipped in `data/transaction_list_*.txt`; all intermediate values in the worked examples were recomputed from those files and confirmed on the curve.

The frequency section reports an exact enumeration of the class-E density on the test curve, a 1.2×10^9 -signature Monte-Carlo of the generator, and an audit of the shipped reports against the current classifier.

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1 Scope, and what is *not* claimed

Every quantity in this document is derived from three public numbers: the message hash z and the signature pair (r, s) . The classification names situations in which the secret intermediate $s_{zk} = zk^{-1}$ happens to coincide with a value computable from those three numbers alone. When it does, the private key follows in closed form and can be confirmed on the curve.

Three points fix the scale of the result, and they must be read before any formula below.

- 1. The class labels are not computable from public data.** The conditions that define classes A, B, C and D are written in terms of s_{zk} and s_{rxk} , both of which contain k^{-1} and are secret. What a holder of (z, r, s) can do is compute a small *candidate set* — one value for class A, at most two for class B, at most two for class E — derive a private-key candidate from each, and test it on the curve. The class label is then only a name for “this candidate turned out to be correct”. It is not a filter that can be applied in advance.
- 2. The hit rate is the rate of guessing k^{-1} at random.** Section 6 establishes $\Pr[A] \approx 1/(4n)$ and $\Pr[E] \approx 4/(3n)$ on a curve of prime order n . On the toy curve secp17k1 ($n = 99\,667$) that is $2.5 \cdot 10^{-6}$ and $1.3 \cdot 10^{-5}$ per signature, which is why the test data contains hundreds of them. On secp256k1 the same quantities are $2.2 \cdot 10^{-78}$ and $1.2 \cdot 10^{-77}$.
- 3. The secp256k1 rows in the data were constructed, not found.** The file `transaction_list_20260812223812.txt` contains 50 class-A and 50 class-E signatures on real secp256k1. They were built by `class_forge.py`, which chooses k and the two additive parts and then *solves* for z and x . That requires already knowing the private key and being free to pick z , so those rows are synthetic test vectors for the recovery path, not signatures anybody could have obtained from a signer.

Nothing in this document constitutes an attack on ECDSA, on secp256k1, or on any deployed system. The algebra is elementary and the recovery step $x \equiv z(s - s_{zk})(rs_{zk})^{-1}$ is the standard consequence of knowing the nonce; the only content of the taxonomy is a census of how often a particular guess for s_{zk} lands on the truth.

2 Preliminaries

2.1 Curve parameters

Two curves appear in the data. Both are of the form $y^2 = x^3 + b$ over $\mathbb{F}_p$, both have prime group order n and cofactor 1.

	secp17k1 (<i>test</i>)	secp256k1 (<i>legacy</i>)
p	100 003	$2^{256} - 2^{32} - 977$
b	2	7
n	99 667	$\approx 1.1579 \cdot 10^{77}$
G	(20 002, 57 568)	standard base point
z	random, <i>not</i> a message hash	double SHA-256 of a Bitcoin sighash preimage
k	random	RFC 6979, deterministic

Because *test*-mode z is drawn at random rather than hashed from a message, test-mode triples are valid (z, r, s) triples satisfying every algebraic relation below, but they are not signatures a verifier would accept against a particular message. Everything in this document is a statement about the algebra of the triples, and is therefore unaffected.

2.2 The signature equations

Let G have prime order n , let $x \in [1, n-1]$ be the private key and $Q = xG$ the public key. A signature on z with nonce $k \in [1, n-1]$ is

$$r \equiv (kG)_x \pmod{n}, \quad s \equiv k^{-1}(z + rx) \pmod{n}. \quad (1)$$

All residues are taken as their canonical integer representatives in $[0, n-1]$; this matters, and Section 3.4 says why.

2.3 The additive split

Lemma 2.1 (Additive split). *Define*

$$s_{zk} \equiv z k^{-1} \pmod{n}, \quad s_{rxk} \equiv r x k^{-1} \pmod{n}. \quad (2)$$

Then $s \equiv s_{zk} + s_{rxk} \pmod{n}$.

Proof. Immediate from (1): $s \equiv k^{-1}(z + rx) \equiv z k^{-1} + r x k^{-1}$. $\square$

Both parts contain k^{-1} , so both are secret. The entire taxonomy is a study of when one of them is nevertheless pinned down by (z, r, s) .

Definition 2.2 (Carry and the integer sum). Since $s_{zk}, s_{rxk} \in [0, n-1]$, their *integer* sum lies in $[0, 2n-2]$ and is congruent to s modulo n . Hence it equals exactly one of s or $s + n$. Write

$$c = \begin{cases} 1 & \text{if } s_{zk} + s_{rxk} > s, \\ 0 & \text{otherwise,} \end{cases} \quad S = s + c n = s_{zk} + s_{rxk} \pmod{n}.$$

The two branches are labelled **E1** ($c = 0, S = s$) and **E2** ($c = 1, S = s + n$) in the reports.

2.4 The master recovery lemma

Lemma 2.3 (Key recovery from s_{zk}). *Let (z, r, s) be a signature and suppose s_{zk} is known, with $\gcd(r s_{zk}, n) = 1$. Then*

$$x \equiv z(s - s_{zk})(r s_{zk})^{-1} \pmod{n} \quad (3)$$

Proof. From (2), $k^{-1} \equiv s_{zk} z^{-1}$. By Lemma 2.1, $s_{rxk} \equiv s - s_{zk}$. Substituting into $s_{rxk} \equiv r x k^{-1}$ gives $s - s_{zk} \equiv r x s_{zk} z^{-1}$, and solving for x yields (3). Every factor on the right is public. $\square$

Equation (3) is the single primitive behind classes A, B and E. The three classes differ only in *which* value is substituted for s_{zk} .

2.5 The public confirmation test

A candidate produced by (3) is a guess and must be checked. The check also uses public data only.

Lemma 2.4 (On-curve confirmation). *Let $\hat{x} \in [1, n-1]$ be a candidate. Put $\hat{k} \equiv (z + r\hat{x}) s^{-1} \pmod{n}$ and accept $\hat{x}$ iff*

$$\hat{k} \not\equiv 0 \quad \text{and} \quad (\hat{k}G)_x \pmod{n} = r.$$

The true key always passes. A candidate drawn independently of the signature passes with probability $\approx 2/n$.

Sketch. If $\hat{x} = x$ then $\hat{k} \equiv (z + rx)s^{-1} \equiv k$ by (1), and $(kG)_x \bmod n = r$ by construction. Conversely, for $\hat{x}$ independent of the signature, $\hat{k}$ is effectively uniform and $(\hat{k}G)_x \bmod n$ hits the fixed target r for about two of the $n - 1$ possible $\hat{k}$, since a given x -coordinate corresponds to the two points $\pm y$. $\square$

Remark 2.5. This is why the reports distinguish *Recovery Attempts* from *Recovery Verified*. A rejected candidate is counted and discarded; it is never printed in the `x_recovered` column. In the audit of Section 7 not a single false positive survived the test across 1 566 attempts.

2.6 Public and secret quantities

Public	z, r, s , and anything computed from them: s_{zr}, d, a, S, u, v , the quadratic and its roots
Secret	$x, k, k^{-1}, s_{zk}, s_{rxk}, m_1, m_2, f$, the carry c , the level N

In the report files a trailing asterisk on a column header marks a secret column, printed for research validation only.

3 The auxiliary quantities

3.1 Definition

Definition 3.1 (s_{zr}, d, a). Set

$$s_{zr} \equiv zr \pmod{n}.$$

If $s_{zr} > s$ as integers, set

$$d = s_{zr} - s, \quad a = s \bmod d.$$

The value a is called *valid* when $0 < a \neq s$; the reports count these under *Transactions With Valid A*. Classes A, B, C and D are defined only when a is valid; when it is not, the signature is simply unclassified in that dimension.

Proposition 3.2 (When a is valid). *Given $s_{zr} > s$, the value $a = s \bmod d$ fails to be valid in exactly two ways: $a = 0$ iff $d \mid s$, and $a = s$ iff $s < d$. The guard $a \neq s_{zr}$ that appears in some earlier revisions is vacuous, because $a < d < s_{zr}$ always.*

Empirically $\Pr[a \text{ valid}] \approx 1/4$: the shipped 10^8 -signature run reports 24 985 980 valid- a signatures (24.99%), and the Monte-Carlo of Section 6 gives 24.99%.

3.2 Why a carries information at all

Lemma 3.3. *Let $P \equiv kr \pmod{n}$. Then $s_{zr} \equiv s_{zk} \cdot P \pmod{n}$.*

Proof. $s_{zk} \cdot P \equiv (zk^{-1})(kr) \equiv zr \equiv s_{zr}$. $\square$

So s_{zr} is a multiplicative image of the very quantity the classifier is hunting, scaled by a factor P that is fixed the moment k is chosen. This is the structural reason a formula in (s, s_{zr}) can occasionally reproduce s_{zk} — and also the reason it cannot be steered: the map $k \mapsto k(kG)_x$ behaves as a random oracle, so P is not a controllable parameter. Lemma 3.3 is the pivot of the whole taxonomy.

3.3 The two ratios

Definition 3.4 (m_1, m_2 , level, remainder). With a valid, define the *rational* numbers

$$m_1 = \frac{s_{zk}}{a} \in \mathbb{Q}, \quad m_2 = \frac{s + s_{zr}}{s_{rxk}} \in \mathbb{Q},$$

computed over $\mathbb{Z}$, not modulo n . Write further

$$N = \lfloor s_{zk}/a \rfloor \quad (\text{the level}), \quad w = s_{zk} \bmod a \quad (\text{the remainder}), \quad f = s_{zk}/a$$

so that $s_{zk} = aN + w$ with $0 \leq w < a$, and $m_1 \in \mathbb{Z}$ iff $w = 0$.

Remark 3.5. m_1 and m_2 are secret: m_1 contains s_{zk} and m_2 contains s_{rxk} . They are the classifier's instruments, not an attacker's. The report column f is m_1 printed to 20 (test) or 80 (legacy) exact decimal places.

3.4 Representative dependence

The tests $s_{zr} > s$, $m_1 > 1$, $S \bmod 4$ and $|s_{zk} - s_{rxk}|$ all treat elements of $\mathbb{Z}_n$ as *integers*. Since n is odd, s and $s + n$ are the same group element with opposite parity, and class E exploits exactly this by choosing $S \in \{s, s + n\}$. Consequently every statistic in this document describes the chosen canonical representatives, not the group $\mathbb{Z}_n$. Rescaling the representatives changes the classification.

4 The taxonomy

4.1 Master table

Classes A–D partition the signatures for which a is valid. Class E is defined independently of a and may overlap any of them — the shipped 10^8 run records 173 class-E signatures that also have a valid a .

Class	Identification condition (research form)	Public candidate set for s_{zk}
A	$m_1 = 1$, i.e. $s_{zk} = a$	$\{a\}$
B	$m_1, m_2 \in \mathbb{Z}$ and $m_1 = m_2$	$\{a\mu : \mu \text{ root of } a\mu^2 - s\mu + (s + s_{zr}) \equiv 0\}$
C	$m_1 \in \mathbb{Z}, m_1 > 1$	$\emptyset$
D	$m_1 \notin \mathbb{Z}$	$\emptyset$
E	$s_{zk} + s_{rxk} = S$ and $ s_{zk} - s_{rxk} = \lfloor S/2 \rfloor + 1$, $S \in \{s, s + n\}, S \equiv 1, 2 \pmod{4}$	$\{u, v\}$ with $u = \lfloor (3S + 2)/4 \rfloor$, $v = \lfloor (S - 1)/4 \rfloor$

The unifying statement. Each of the three recoverable classes pins down the *unordered pair* $\{s_{zk}, s_{rxk}\}$, not s_{zk} itself:

- class A gives $\{a, s - a\}$;
- class B gives $\{a\mu, a\mu'\}$ where μ, μ' are the two roots of the quadratic, and $a\mu' \equiv s - a\mu$ by Vieta (Theorem 4.4);
- class E gives $\{u, v\}$ with $u + v = S$.

Which member of the pair is s_{zk} is *not* determined, so the recovery procedure tries both and lets Lemma 2.4 decide. This is why the shipped report records 2 attempts and 1 verification per class-E signature, and exactly 670 rejections for 670 class-E signatures.

4.2 Class A

Definition 4.1 (Class A). A signature with valid a is of class A iff $s_{zk} = a$, equivalently $m_1 = 1$.

Theorem 4.2 (Structure of class A). With $d = s_{zr} - s$ and carry c as in Definition 2.2, class A holds iff

$$d \mid (s_{rxk} - cn) \quad \text{and} \quad s_{zk} < d.$$

In the no-carry case $c = 0$ this reads: d divides s_{rxk} , and s_{zk} is smaller than d .

Proof. By definition $a = s \bmod d$, so $a = s_{zk}$ iff $d \mid (s - s_{zk})$ and $0 \leq s_{zk} < d$. Since $S = s_{zk} + s_{rxk} = s + cn$ in $\mathbb{Z}$, we have $s - s_{zk} = s_{rxk} - cn$. $\square$

Corollary 4.3 (Constructibility). To build a class-A signature one chooses s_{zk} , then any divisor $d \mid (s_{zr} - s_{zk})$ with $s_{zk} < d$; with $s_{zk} = 1$ this needs only $d = 2$ or $d = 3$, which is why `class_forge.py` can produce class A on `secp256k1` after trying about two nonces. Construction requires knowing x and k and being free to choose z ; it is not an attack.

Class A recovery. Substitute $s_{zk} = a$ into (3):

$$\hat{x}_A \equiv z(s - a)(ra)^{-1} \pmod{n}, \quad a = s \bmod (s_{zr} - s). \quad (4)$$

One candidate, one attempt. Confirm with Lemma 2.4.

4.3 Class B

Theorem 4.4 (The class-B quadratic). Suppose $m_1 = m_2 = \mu \in \mathbb{Z}$. Then

$$a\mu^2 - s\mu + (s + s_{zr}) \equiv 0 \pmod{n}. \quad (5)$$

Its roots are

$$\mu \equiv (s \pm \sqrt{\Delta})(2a)^{-1} \pmod{n}, \quad \Delta = s^2 - 4a(s + s_{zr}) \bmod n, \quad (6)$$

computable whenever Δ is a quadratic residue modulo the prime n . Moreover, if μ, μ' are the two roots then

$$a\mu' \equiv s - a\mu \equiv s_{rxk} \pmod{n},$$

so the root pair returns exactly the unordered pair $\{s_{zk}, s_{rxk}\}$.

Proof. $m_1 = \mu$ gives $s_{zk} = a\mu$; $m_2 = \mu$ gives $s_{rxk} = (s + s_{zr})/\mu$. Lemma 2.1 gives $a\mu + (s + s_{zr})\mu^{-1} \equiv s$; multiplying by μ yields (5), and (6) is the quadratic formula over the field $\mathbb{Z}_n$. For the last claim, Vieta gives $\mu + \mu' \equiv s a^{-1}$, hence $a\mu' \equiv s - a\mu \equiv s - s_{zk} \equiv s_{rxk}$. $\square$

Class B recovery. For each root μ of (5) put $s_{zk}^{(\mu)} = a\mu \bmod n$ and

$$\hat{x}_B \equiv z(s - a\mu)(ra\mu)^{-1} \pmod{n}. \quad (7)$$

At most two candidates, at most two attempts.

Remark 4.5 (Why class B is essentially unreachable on secp256k1). $m_1 = m_2$ pins the ratio $s_{zr} : s_{zk}$ to a fixed rational. By Lemma 3.3 that ratio is $P = kr \bmod n$, which cannot be steered. On the toy curve all $\approx 10^5$ nonces can be enumerated until one hits the target; on secp256k1 the same search costs 2^{-256} per nonce, and `class_forge.py` refuses the request with an explicit `ClassConstructionError` rather than looping.

4.4 Class C: the level obstruction

Definition 4.6 (Class C). A signature with valid a is of class C iff $m_1 \in \mathbb{Z}$ and $m_1 > 1$ (and it is not class B).

Here $s_{zk} = a m_1$ with a public but m_1 an unknown positive integer. Nothing in (z, r, s) determines m_1 : a is a function of (s, s_{zr}) , while m_1 absorbs the whole of the unknown s_{zk} . The public candidate set is therefore empty and no recovery formula exists. Exhaustive search over $m_1 \in [2, n/a]$ is possible in principle but is a search of size n/a , i.e. generically $\Theta(n)$.

4.5 Class D: no integral level

Definition 4.7 (Class D). A signature with valid a is of class D iff $m_1 \notin \mathbb{Z}$, i.e. $w = s_{zk} \bmod a > 0$. The reports label such rows DN with $N = \lfloor s_{zk}/a \rfloor$.

Class D is the generic case: it accounts for 24 929 792 of the 24 985 980 valid- a signatures in the shipped run, i.e. 99.78% of them. As with class C, s_{zk} is not determined and there is no recovery formula.

4.5.1 Level windows

If the level N is supplied from outside — it is *not* recoverable from public data — then s_{zk} is confined to

$$s_{zk} \in W = [aN, a(N+1) - 1], \quad |W| = a. \quad (8)$$

`find_common_private_key.py` enumerates W for several transactions believed to share a key, maps each trial s_{zk} through (3), and intersects the resulting candidate sets. The cost is $\sum_i |W_i|$ rather than the Cartesian product $\prod_i |W_i|$. For κ transactions the expected number of *coincidental* survivors before verification is

$$\mathbb{E}[\text{impostors}] \approx \frac{\prod_{i=1}^{\kappa} |W_i|}{n^{\kappa-1}}, \quad (9)$$

which in the bundled four-transaction demo equals 1.016 — so the raw intersection proves nothing and the on-curve test of Lemma 2.4 is what makes the answer meaningful. On secp256k1 a single window has size $\approx a \sim n$ and the module refuses to run.

4.5.2 HYP-001, and an exact reduction

The repository ships one example hypothesis for class D. With $N = \lfloor s_{zk}/a \rfloor$ and $w = s_{zk} \bmod a$, and under the guards $N > 1$, $s > s_{zk}$, s_{zk} even and $w > 0$, HYP-001 fires when

$$Nw - ((Na \bmod w) - ((N+1)a \bmod w)) = a. \quad (10)$$

Theorem 4.8 (Exact reduction of HYP-001). Let $t = a \bmod w$ and

$$\delta = \begin{cases} 1 & \text{if } ((Nt) \bmod w) + t \geq w, \\ 0 & \text{otherwise.} \end{cases}$$

Then (10) holds if and only if

$$\lfloor a/w \rfloor = N - \delta.$$

Proof. Write $a = qw + t$ with $q = \lfloor a/w \rfloor$ and $0 \leq t < w$. Then $Na \bmod w = Nt \bmod w$ and $(N+1)a \bmod w = (N+1)t \bmod w$. Since $(N+1)t \equiv Nt + t$, we have $(N+1)t \bmod w = (Nt \bmod w) + t - \delta w$ with δ as stated. Hence

$$(Na \bmod w) - ((N+1)a \bmod w) = -t + \delta w,$$

and the left side of (10) equals $Nw + t - \delta w = w(N - \delta) + t$. Setting this equal to $a = qw + t$ and cancelling t gives $q = N - \delta$. $\square$

Remark 4.9 (Correction to the repository documentation). The README states that (10) “reduces to $\lfloor a/w \rfloor \in \{N, N - 1\}$ ”. Theorem 4.8 shows that this is a *necessary* condition only: which of the two values must occur is determined by δ . A direct test over $3 \cdot 10^5$ random triples (N, a, w) found 380 firings of (10) and 912 triples satisfying $\lfloor a/w \rfloor \in \{N, N - 1\}$; all 380 firings satisfied the weaker condition and 532 triples satisfied it without (10) holding. The exact characterisation of Theorem 4.8 matched (10) on $5 \cdot 10^5$ random triples with zero disagreements.

Theorem 4.8 also explains the scale-invariance reported in the repository: the condition constrains the first two partial quotients of the continued-fraction expansion of s_{zk}/a , whose distribution is governed by the Gauss–Kuzmin law and is therefore independent of the size of n . It is a property of the ratio s_{zk}/a , not of ECDSA. Measured frequencies appear in Section 6.

4.6 Class E: the half-difference split

Definition 4.10 (Class E). Let $S = s_{zk} + s_{rxk} \in \{s, s + n\}$ be the integer sum of Definition 2.2. The signature is of class E iff

$$|s_{zk} - s_{rxk}| = \lfloor S/2 \rfloor + 1. \quad (11)$$

Theorem 4.11 (Integrality and closed form). *The system*

$$u + v = S, \quad u - v = \lfloor S/2 \rfloor + 1, \quad u, v \in \mathbb{Z}$$

is solvable if and only if $S \equiv 1$ or $S \equiv 2 \pmod{4}$, in which case the solution is unique and given by

$$u = \left\lfloor \frac{3S+2}{4} \right\rfloor, \quad v = S - u = \left\lfloor \frac{S-1}{4} \right\rfloor. \quad (12)$$

Moreover

$$u - 3v = S \bmod 4 \in \{1, 2\}. \quad (13)$$

Proof. Write $2u = S + \lfloor S/2 \rfloor + 1$. If $S = 2m$ is even, $2u = 3m + 1$, which is even iff m is odd, i.e. $S \equiv 2 \pmod{4}$, and then $u = (3S + 2)/4$. If $S = 2m + 1$ is odd, $\lfloor S/2 \rfloor = m$ and $2u = 3m + 2$, even iff m is even, i.e. $S \equiv 1 \pmod{4}$, and then $u = (3S + 1)/4$. In both cases $u = \lfloor (3S + 2)/4 \rfloor$, since for $S \equiv 1$ the floor removes the surplus $1/4$. Subtracting, $v = S - u$ equals $(S - 2)/4$ and $(S - 1)/4$ respectively, both of which equal $\lfloor (S - 1)/4 \rfloor$. For $S \equiv 0$ or $3 \pmod{4}$ the parity requirement fails and no integer solution exists. Finally, $u - 3v$ equals 2 in the first case and 1 in the second, matching $S \bmod 4$. $\square$

Remark 4.12. Identity (13) gives class E an intuitive reading: the additive split of S is in ratio 3 : 1 up to an additive slack of 1 or 2. Nothing in the definition refers to s_{zk} , a or m_1 , which is why class E is independent of the A–D chain and why it can be constructed on any curve.

Class E recovery. For $S \in \{s, s + n\}$ with $S \equiv 1, 2 \pmod{4}$ compute u, v by (12) and evaluate (3) at both:

$$\hat{x}_E^{(u)} \equiv z(s - u)(ru)^{-1}, \quad \hat{x}_E^{(v)} \equiv z(s - v)(rv)^{-1} \pmod{n}. \quad (14)$$

At most two candidates per branch; exactly one of them is the truth when the branch is the correct one. Since the carry c is secret, in practice one tries the eligible branches among $\{s, s + n\}$.

4.7 The public decision procedure

Collecting the above, the complete public-data algorithm is:

1. Input (z, r, s) and curve parameters. Set $\mathcal{C} \leftarrow \emptyset$.
2. $s_{zr} \leftarrow zr \bmod n$.
3. If $s_{zr} > s$: $d \leftarrow s_{zr} - s$; $a \leftarrow s \bmod d$; if $0 < a \neq s$ then
 - (a) $\mathcal{C} \leftarrow \mathcal{C} \cup \{a\}$ (class-A candidate)
 - (b) $\Delta \leftarrow s^2 - 4a(s + s_{zr}) \bmod n$; for every square root ρ of $\Delta \bmod n$: $\mu \leftarrow (s + \rho)(2a)^{-1}$,
 $\mathcal{C} \leftarrow \mathcal{C} \cup \{a\mu \bmod n\}$ (class-B candidates)
4. For $S \in \{s, s + n\}$ with $S \equiv 1, 2 \pmod{4}$: $u \leftarrow \lfloor (3S + 2)/4 \rfloor$, $v \leftarrow S - u$, $\mathcal{C} \leftarrow \mathcal{C} \cup \{u \bmod n, v \bmod n\}$ (class-E candidates)
5. For each $\sigma \in \mathcal{C}$ with $\gcd(r\sigma, n) = 1$: $\hat{x} \leftarrow z(s - \sigma)(r\sigma)^{-1} \bmod n$; accept $\hat{x}$ iff it passes Lemma 2.4.

$|\mathcal{C}| \leq 7$ always. Classes C and D contribute nothing to $\mathcal{C}$, which is the precise sense in which they are not recoverable. The procedure succeeds with probability $\approx |\mathcal{C}|/n$ per signature.

5 Worked examples

Every number below was recomputed from the shipped report files and re-verified on the curve; none is quoted from the reports without an independent check. Secret columns are shown only to demonstrate that the public formula landed on the truth.

5.1 Class A on secp17k1

Source: transaction_list_20260809003013.txt, first row labelled A. Here $n = 99667$.

Public data.

$$z = 11681, \quad r = 54911, \quad s = 57604.$$

Secret data (for validation only).

$$s_{zk} = 466, \quad s_{rxk} = 57138, \quad x = 41338, \quad k = 24621.$$

Step 1 — auxiliary quantities.

$$s_{zr} = zr \bmod n = 11681 \cdot 54911 \bmod 99667 = 58246,$$

$$s_{zr} > s \quad (58246 > 57604) \Rightarrow d = s_{zr} - s = 642,$$

$$a = s \bmod d = 57604 \bmod 642 = 466 \quad \text{since } 57604 = 89 \cdot 642 + 466.$$

a is valid: $0 < 466 \neq 57604$.

Step 2 — identification. $m_1 = s_{zk}/a = 466/466 = 1$, so the row is class A. Theorem 4.2 is confirmed directly: the carry is $c = 0$ and $s_{rxk} = 57138 = 89 \cdot 642$, so $d \mid s_{rxk}$, while $s_{zk} = 466 < 642 = d$.

Step 3 — recovery by (4).

$$\begin{aligned} z(s - a) \bmod n &= 11681 \cdot (57604 - 466) \bmod 99667 = 58746, \\ ra \bmod n &= 54911 \cdot 466 \bmod 99667 = 73774, \\ (ra)^{-1} \bmod n &= 69601, \\ \hat{x} &= 58746 \cdot 69601 \bmod 99667 = \boxed{41338}. \end{aligned}$$

Step 4 — confirmation (Lemma 2.4).

$$\hat{k} = (z + r\hat{x})s^{-1} \bmod n = 24621, \quad (\hat{k}G)_x \bmod n = 54911 = r \checkmark$$

The recovered key equals the generator's $x = 41338$, and matches the `x_recovered` column of the report. One attempt, one verification.

5.2 Class B on secp17k1

Source: `transaction_list_20260809003013.txt`, the single row labelled B in the whole 10^8 -signature run.

Public data.

$$z = 3481, \quad r = 33385, \quad s = 1197.$$

Secret data. $s_{zk} = 665$, $s_{rxk} = 532$, $x = 78607$, $k = 12295$.

Step 1 — auxiliary quantities.

$$s_{zr} = 1463, \quad d = s_{zr} - s = 266, \quad a = s \bmod d = 133 \quad (1197 = 4 \cdot 266 + 133).$$

Step 2 — identification.

$$m_1 = \frac{s_{zk}}{a} = \frac{665}{133} = 5, \quad m_2 = \frac{s + s_{zr}}{s_{rxk}} = \frac{1197 + 1463}{532} = 5.$$

Both are integers and $m_1 = m_2 = 5$, so the row is class B.

Step 3 — the quadratic (5).

$$\begin{aligned} \Delta &= s^2 - 4a(s + s_{zr}) \bmod n = 1197^2 - 4 \cdot 133 \cdot (1197 + 1463) \bmod 99667 = 17689, \\ \sqrt{\Delta} &\in \{133, 99534\} \quad (\text{note } 133^2 = 17689), \\ (2a)^{-1} \bmod n &= 46836, \\ \mu &\equiv (s \pm \sqrt{\Delta})(2a)^{-1} \in \{5, 4\}. \end{aligned}$$

The two roots give the candidate pair $a\mu \in \{665, 532\}$, which is exactly $\{s_{zk}, s_{rxk}\} = \{665, 532\}$ — the Vieta statement of Theorem 4.4 in action.

Step 4 — recovery by (7), both roots.

Root $\mu = 5$, s_{zk} -guess 665:

$$\hat{x} = 57886 \cdot 76581 \bmod 99667 = \boxed{78607}, \quad \hat{k} = 12295, \quad (\hat{k}G)_x = 33385 = r \checkmark$$

Root $\mu = 4$, s_{zk} -guess 532:

$$\hat{x} = 41844, \quad \hat{k} = 90119, \quad (\hat{k}G)_x = 16128 \neq r = 33385 \text{ rejected.}$$

Two attempts, one verification; the surviving candidate equals the true key 78607.

5.3 Class C on secp17k1 — nothing is recoverable

Source: transaction_list_20260809003013.txt, first row labelled C.

$$z = 80705, \quad r = 61603, \quad s = 80812, \quad s_{zr} = 80821, \quad d = 9, \quad a = 1.$$

Here $s = 8979 \cdot 9 + 1$, so $a = 1$ and

$$m_1 = \frac{s_{zk}}{a} = \frac{95037}{1} = 95037 \in \mathbb{Z}, \quad m_1 > 1,$$

which is class C. The identity $s_{zk} = a m_1$ holds, but with $a = 1$ it says only $s_{zk} = m_1$: the public quantity a carries no information and the multiplier absorbs the entire secret. The public candidate set is empty; the report's `x_recovered` column is `-`.

Exhaustive search would mean trying all $m_1 \in [2, n/a] = [2, 99667]$, i.e. the full group. For reference the true values are $s_{zk} = 95037$, $s_{rxk} = 85442$, $x = 92463$ — none of which is deducible from the three public numbers above.

5.4 Class D on secp17k1 — nothing is recoverable

Source: transaction_list_20260809003013.txt, first row, labelled D3.

$$z = 8144, \quad r = 44375, \quad s = 57116, \quad s_{zr} = 97125, \quad d = 40009, \quad a = 17107.$$

$$m_1 = \frac{s_{zk}}{a} = \frac{66522}{17107} \notin \mathbb{Z}, \quad N = \lfloor s_{zk}/a \rfloor = 3, \quad w = s_{zk} \bmod a = 15201.$$

The report prints $f = s_{zk}/a$ to 20 exact decimal places as 3.88858362074004793359 (pure integer arithmetic; a double-precision float would already be wrong in the 17th digit).

If the level $N = 3$ were supplied from outside, (8) would confine s_{zk} to

$$W = [aN, a(N+1) - 1] = [51321, 68427], \quad |W| = a = 17107,$$

a window of 17107 candidates out of $n = 99667$. But N is a function of the secret s_{zk} ; it is an input to `find_common_private_key.py`, never an output. From (z, r, s) alone the candidate set is empty.

A class-D row that fires HYP-001

The same file contains rows flagged HYP-001. Taking the first one (labelled D2):

$$s = 49952, \quad s_{zr} = 90013, \quad d = 40061, \quad a = 9891, \quad s_{zk} = 26544, \quad N = 2, \quad w = 6762.$$

Condition (10) evaluates to

$$Nw - ((Na \bmod w) - ((N+1)a \bmod w)) = 2 \cdot 6762 - (6258 - 2625) = 9891 = a \quad \checkmark$$

and Theorem 4.8 reproduces it: $t = a \bmod w = 3129$, $((Nt) \bmod w) + t = 6258 + 3129 = 9387 \geq 6762 = w$, hence $\delta = 1$ and $\lfloor a/w \rfloor = \lfloor 9891/6762 \rfloor = 1 = 2 - 1 = N - \delta \quad \checkmark$.

HYP-001 changes nothing about recoverability: the row remains class D and the `x_recovered` column remains `-`. It is a statement about the continued fraction of s_{zk}/a , both of whose terms are secret.

5.5 Class E1 on secp17k1 (no carry, $S = s$)

Source: transaction_list_20260809003013.txt, first row labelled E1.

$$z = 99119, \quad r = 73199, \quad s = 19582.$$

Secret: $s_{zk} = 14687$, $s_{rxk} = 4895$, $x = 31386$, $k = 49864$.

Step 1 — branch and eligibility. $s_{zk} + s_{rxk} = 14687 + 4895 = 19582 = s$, so $c = 0$ and $S = s = 19582$. Then $S \bmod 4 = 2 \in \{1, 2\}$, so Theorem 4.11 applies.

Step 2 — identification.

$$\lfloor S/2 \rfloor + 1 = 9792, \quad |s_{zk} - s_{rxk}| = |14687 - 4895| = 9792 \checkmark$$

Step 3 — the public pair (12).

$$u = \lfloor \frac{3S+2}{4} \rfloor = \lfloor \frac{3 \cdot 19582 + 2}{4} \rfloor = 14687, \quad v = S - u = 4895, \quad u - 3v = 2 = S \bmod 4 \checkmark$$

Here $u = s_{zk}$ and $v = s_{rxk}$, but that ordering is not knowable in advance.

Step 4 — recovery (14), both members.

$$\begin{aligned} s_{zk} = u = 14687 : \quad \hat{x} &= 8549 \cdot 68473 \bmod 99667 = \boxed{31386}, \quad (\hat{k}G)_x = 73199 = r \checkmark \\ s_{zk} = v = 4895 : \quad \hat{x} &= 77741, \quad (\hat{k}G)_x = 55484 \neq r \text{ rejected} \end{aligned}$$

Two attempts, one verification; 31386 is the generator's key.

5.6 Class E2 on secp17k1 (carry, $S = s + n$)

Source: same file, first row labelled E2. This is the branch that makes the representative-dependence of Section 3.4 concrete.

$$z = 97391, \quad r = 95346, \quad s = 3723.$$

Secret: $s_{zk} = 25847$, $s_{rxk} = 77543$, $x = 29536$, $k = 77433$.

Step 1 — branch. $s_{zk} + s_{rxk} = 25847 + 77543 = 103390$, which exceeds $s = 3723$; hence $c = 1$ and $S = s + n = 3723 + 99667 = 103390$. The other branch is ineligible: $s \bmod 4 = 3 \notin \{1, 2\}$. For $S = s + n$ we have $S \bmod 4 = 2 \checkmark$.

Step 2 — identification.

$$\lfloor S/2 \rfloor + 1 = 51696 = |s_{zk} - s_{rxk}| = 51696 \checkmark$$

Step 3 — the public pair.

$$u = 77543, \quad v = 25847, \quad u - 3v = 2 = S \bmod 4.$$

This time $s_{zk} = v$ is the *smaller* member — the reason both are always tried.

Step 4 — recovery.

$$\begin{aligned} s_{zk} = u = 77543 : \quad \hat{x} &= 3863, \quad (\hat{k}G)_x = 34957 \neq r \text{ rejected} \\ s_{zk} = v = 25847 : \quad \hat{x} &= 22389 \cdot 37791 \bmod 99667 = \boxed{29536}, \quad (\hat{k}G)_x = 95346 = r \checkmark \end{aligned}$$

5.7 Class A on real secp256k1

Source: transaction_list_20260812223812.txt, first row labelled A. These signatures were *constructed* by class_forge.py, not found by sampling: z was solved for, so it is not the hash of any message. The row is nonetheless an honest ECDSA signature and every relation below was verified on the curve.

Public data.

Z	81450868 08902802 52130838 65127905 15057001 11517359 75240539 91466162 19855876 2499
r	57079190 47754951 24819687 29810762 28441484 72895901 02352992 54317235 30626565 15512
S	12508468 19273237 79356594 86354472 80318025 56010228 26811916 91955654 38170650 03437

Secret data (validation only).

s_{zk}	1
s_{rxk}	12508468 19273237 79356594 86354472 80318025 56010228 26811916 91955654 38170650 03436
x	74140579 79830083 19481469 37240142 04636861 34269367 26218208 37604175 26583118 8393
k	81450868 08902802 52130838 65127905 15057001 11517359 75240539 91466162 19855876 2499

Note $k = z$ here: the forge sets $z = s_{zk} \cdot k$ and chose $s_{zk} = 1$.

Step 1 — auxiliary quantities.

$s_{zr} =$	12508468 19273237 79356594 86354472 80318025 56010228 26811916 91955654 38170650 03440
$zr \bmod n$	
$d = s_{zr} - s$	3
$a = s \bmod d$	1
$\lfloor s/d \rfloor$	41694893 97577459 31188649 54514909 34393418 53367427 56039723 06518847 93902166 7812

Indeed $s_{zr} = s + 3$, and $s \equiv 1 \pmod{3}$, so $a = 1$.

Step 2 — identification. $m_1 = s_{zk}/a = 1/1 = 1$: class A. Theorem 4.2 holds with $c = 0$, since $s_{rxk} = s - 1$ is divisible by $d = 3$ and $s_{zk} = 1 < 3$.

Step 3 — recovery by (4). With $a = 1$ the formula collapses to $\hat{x} \equiv z(s - 1)r^{-1} \pmod{n}$:

$z(s -$	14882887 78559854 32535751 59665327 72412826 09158577 02971124 70818102 67721717 25895
$a) \bmod n$	
$ra \bmod n$	57079190 47754951 24819687 29810762 28441484 72895901 02352992 54317235 30626565 15512
$(ra)^{-1} \bmod n$	72009562 40259788 71822971 56000514 13382188 02209984 31678810 79846225 42572386 86715
$\hat{x}$	74140579 79830083 19481469 37240142 04636861 34269367 26218208 37604175 26583118 8393

Step 4 — confirmation.

$\hat{k} =$	81450868 08902802 52130838 65127905 15057001 11517359 75240539 91466162 19855876 2499
$(z + r\hat{x})s^{-1}$	
$(\hat{k}G)_x \bmod n$	57079190 47754951 24819687 29810762 28441484 72895901 02352992 54317235 30626565 15512
r	57079190 47754951 24819687 29810762 28441484 72895901 02352992 54317235 30626565 15512

$(\hat{k}G)_x \bmod n = r$, and $\hat{x}$ equals the generator's private key. One attempt, one verification.

5.8 Class E1 on real secp256k1 ($S \equiv 1 \pmod{4}$)

Source: transaction_list_20260812223812.txt, first row labelled E1. This is the family that the repository's earlier detector discarded entirely (Section 7).

Public data.

Z	11268773 77676412 20182337 61671242 48798274 83966732 58728779 22117527 66804547 150001
r	50335723 62212775 45716919 42099499 71653367 43170162 84055158 97076885 22978187 64226

S 72051455 06168836 01618552 47968643 26367087 83329625 32071822 94119907 14544928 80269

Secret data.

s_{zk} 18012863 76542209 00404638 11992160 81591771 95832406 33017955 73529976 78636232 20067
 s_{rxk} 54038591 29626627 01213914 35976482 44775315 87497218 99053867 20589930 35908696 60202
 x 98309429 03385047 78042801 29297504 52218902 92853422 88652556 98132226 08909335 57242

Step 1 — branch and eligibility. $s_{zk} + s_{rxk} = s$ exactly, so $c = 0$ and $S = s$, with $S \bmod 4 = 1$.

Step 2 — identification.

$\lfloor S/2 \rfloor + 1$ 36025727 53084418 00809276 23984321 63183543 91664812 66035911 47059953 57272464 40135
 $|s_{zk} - s_{rxk}|$ 36025727 53084418 00809276 23984321 63183543 91664812 66035911 47059953 57272464 40135

Step 3 — the public pair (12).

$u =$ 54038591 29626627 01213914 35976482 44775315 87497218 99053867 20589930 35908696 60202
 $\lfloor (3S + 2)/4 \rfloor$
 $v =$ 18012863 76542209 00404638 11992160 81591771 95832406 33017955 73529976 78636232 20067
 $\lfloor (S - 1)/4 \rfloor$
 $u - 3v$ 1 = $S \bmod 4$

Step 4 — recovery, both members. The candidate $s_{zk} = u$ is rejected: it yields $\hat{x}$ with $(\hat{k}G)_x \neq r$. The candidate $s_{zk} = v$ succeeds:

$s_{zk} = v$ 18012863 76542209 00404638 11992160 81591771 95832406 33017955 73529976 78636232 20067
 $z(s -$ 10939111 18459217 86553741 12003822 19009002 37312688 02801387 04233401 57696315 037614
 $v) \bmod n$
 $(rv)^{-1} \bmod n$ 57500374 60391609 80620305 90440258 21991092 21878800 25883856 11365175 06848736 24780
 $\hat{x}$ 98309429 03385047 78042801 29297504 52218902 92853422 88652556 98132226 08909335 57242
 $\hat{k}$ 11304164 17708377 68620733 75239234 82620422 75879550 85438601 39207706 79369641 808167
 $(\hat{k}G)_x \bmod n$ 50335723 62212775 45716919 42099499 71653367 43170162 84055158 97076885 22978187 64226

$\hat{x}$ equals the true x of the row. Two attempts, one verification.

5.9 Class E2 on real secp256k1 (carry branch)

Source: same file, first row labelled E2.

Public data.

z 84509321 30679407 48985591 87787221 27000258 72833510 16819540 04417130 31634789 76749
 r 85576404 99117476 59035787 06601795 74135573 88971266 23850946 47629191 62775627 28325
 s 77201038 16614255 72861140 75001117 23789173 47266960 40868876 73647689 19990878 1740

Step 1 — branch. The no-carry branch is ineligible, $s \bmod 4 = 0$. The carry branch gives

$S = s + n$ 12351219 30539304 51152182 39250879 96316420 11036948 67899127 02788108 30718070 276077
 $S \bmod 4$ 1

Step 2 — identification.

$\lfloor S/2 \rfloor + 1$ 61756096 52696522 55760911 96254399 81582100 55184743 39495635 13940541 53590351 38039
 $|s_{zk} - s_{rxk}|$ 61756096 52696522 55760911 96254399 81582100 55184743 39495635 13940541 53590351 38039

Step 3 — the public pair, and recovery.

u 92634144 79044783 83641367 94381599 72373150 82777115 09243452 70910812 30385527 07058
 v 30878048 26348261 27880455 98127199 90791050 27592371 69747817 56970270 76795175 69019
 $u - 3v$ 1
 $\hat{x}$ from v 55314219 87078954 99857298 90156745 34226771 60028719 43901624 00522834 99325621 69296

$(\hat{k}G)_x \bmod n$ 85576404 99117476 59035787 06601795 74135573 88971266 23850946 47629191 62775627 28325
 r 85576404 99117476 59035787 06601795 74135573 88971266 23850946 47629191 62775627 28325

The candidate built from u fails the on-curve test; the one built from v reproduces the row's private key.

5.10 Summary of the worked examples

Curve	Class	Candidates	Attempts	Verified	Rejected	Recovered key
secp17k1	A	$\{a\}$	1	1	0	matches x^*
secp17k1	B	$\{a\mu, a\mu'\}$	2	1	1	matches x^*
secp17k1	C	$\emptyset$	0	0	0	—
secp17k1	D	$\emptyset$	0	0	0	—
secp17k1	E1	$\{u, v\}$	2	1	1	matches x^*
secp17k1	E2	$\{u, v\}$	2	1	1	matches x^*
secp256k1	A	$\{a\}$	1	1	0	matches x^*
secp256k1	E1	$\{u, v\}$	2	1	1	matches x^*
secp256k1	E2	$\{u, v\}$	2	1	1	matches x^*

Across the entire shipped data set the same pattern holds exactly: 1 566 attempts produced 896 verifications and 670 rejections, with zero false positives surviving Lemma 2.4.

6 How often each class occurs

6.1 An exact census for class E

Class E depends only on the pair (s_{zk}, s_{rxk}) , so its density can be counted exactly rather than estimated. Model s_{zk} and s_{rxk} as independent and uniform on $[1, n-1]$; every such pair determines s , the carry c and S . Enumerating all $(n-1)^2 = 9\,933\,311\,556$ pairs on secp17k1 gives the following counts.

Detector	hits	E1	E2	rate
$S \equiv 1, 2 \pmod{4}$ (current code)	132 884	99 664	33 220	$1.338 \cdot 10^{-5}$
$S \equiv 2 \pmod{4}$ only (earlier revision)	66 442	49 832	16 610	$6.689 \cdot 10^{-6}$

Theorem 6.1 (Class-E density). *Under the uniform model, on a curve of prime order n ,*

$$\Pr[\text{class E}] = \frac{\mathcal{N}_1 + \mathcal{N}_2}{(n-1)^2} \approx \frac{4}{3n}, \quad \text{where}$$

$$\mathcal{N}_1 = 2\#\{s : s \equiv 1, 2 \pmod{4}\} \quad (\text{branch E1}),$$

$$\mathcal{N}_2 = 2\#\{s : s < \frac{n-2}{3}, s+n \equiv 1, 2 \pmod{4}\} \quad (\text{branch E2}).$$

Restricting to $S \equiv 2 \pmod{4}$ halves each term, giving $2/(3n)$.

Proof sketch. For branch E1 ($c = 0$) the pair (u, v) of Theorem 4.11 always satisfies $0 < v < u < s$, so both assignments $s_{zk} \in \{u, v\}$ are admissible and each eligible s contributes 2 hits; eligibility has density $1/2$. For branch E2 ($c = 1$) admissibility requires $s_{zk} > s$ and $s_{rxk} < n$, i.e. $v > s$ and $u < n$; both reduce to $s < (n-2)/3$. Hence branch E2 contributes 2 hits for the eligible s below $n/3$, a density of $\frac{1}{2} \cdot \frac{1}{3}$. Summing, $\Pr \approx \frac{1}{n^2}(n + \frac{n}{3}) = \frac{4}{3n}$. The exact enumeration above reproduces this: $132\,884/9\,933\,311\,556 = 1.338 \cdot 10^{-5}$ against $4/(3 \cdot 99\,667) = 1.338 \cdot 10^{-5}$. $\square$

Corollary 6.2. *Exactly half of all class-E signatures have $S \equiv 1 \pmod{4}$: the two rows of the table differ by the factor 2, since $66\,442 \cdot 2 = 132\,884 = 132\,884$. The E1:E2 ratio is 3 : 1 in both rows ($99\,664 : 33\,220$).*

6.2 Monte-Carlo of the generator

The exact census assumes independence. To check it against the real pipeline, the generator was simulated faithfully — the full table $k \mapsto (kG)_x$ on secp17k1, 5000 distinct keys, z and k drawn from $[1000, n - 2]$ exactly as `hash_message` and `_k_candidates` do in test mode — over $1.2 \cdot 10^9$ signatures.

Quantity	shipped report (10^8)	Monte-Carlo (10^8)	rate
valid a	24 985 980	24 991 156	0.2499
class A	224	277	$2.770 \cdot 10^{-6}$
class B	1	0	—
class C	55 964	55 623	$5.562 \cdot 10^{-4}$
class D	24 929 792	24 935 256	0.2494
class E (earlier detector)	670	683	$6.830 \cdot 10^{-6}$
of which E1	519	510	
of which E2	151	173	
class E (current detector)	—	1 379	$1.379 \cdot 10^{-5}$
E overlapping valid a	173	160	
HYP-001	300 744	300 482	0.00300

Agreement is close everywhere. Class A is the one entry with visible tension: 224 observed against ≈ 277 expected, a gap of about 2 standard deviations for Poisson counts of this size — noticeable, not significant.

6.3 Rare-class rates from a larger run

Extending the same simulation to $1.2 \cdot 10^9$ signatures gives

$$\Pr[A] = \frac{2987}{1.2 \cdot 10^9} = 2.489 \cdot 10^{-6} = \frac{0.248}{n} \approx \frac{1}{4n},$$

which is the intuitive value: class A needs a to be valid (probability $\approx 1/4$) and then s_{zk} to land on the single value a (probability $\approx 1/n$).

Class B was observed **zero** times in $1.2 \cdot 10^9$ signatures, giving a one-sided 95% upper bound of $3/1.2 \cdot 10^9 \approx 2.500 \cdot 10^{-9}$ per signature. The single class-B row in the shipped 10^8 run is therefore a genuine but fortunate event: at that rate the probability of seeing at least one in 10^8 signatures is about 22%. Class B is too rare for its frequency to be estimated reliably from either run, and the repository’s own figure of $1.46 \cdot 10^{-10}$ comes from a third run of $6.83 \cdot 10^9$ signatures. All that can be said with confidence is $\Pr[B] \lesssim 10^{-9}$.

6.4 HYP-001 frequency

Over the shipped run HYP-001 fired 300 744 times among 24 929 792 class-D signatures, i.e. 1.206%; the large Monte-Carlo gives 1.207% of class-D signatures and 6.719% once the guards $N > 1$, $s > s_{zk}$, s_{zk} even, $w > 0$ are imposed as the denominator. The repository’s README quotes a null-model rate of 0.50%; the measured class-D rate is roughly twice that, so the null model and the observed population are not describing the same conditioning. Theorem 4.8 explains why the rate is scale-invariant without explaining the level: the condition fixes $\lfloor a/w \rfloor$ relative to N , a statement about two partial quotients of a continued fraction, whose law (Gauss–Kuzmin) does not depend on n .

6.5 Extrapolation to secp256k1

Theorem 6.1 and the measured $\Pr[A] \approx 1/(4n)$ carry over verbatim, because neither depends on anything but the group order.

	secp17k1 ($n = 99\,667$)	secp256k1
$\Pr[A] \approx 1/(4n)$	$2.508 \cdot 10^{-6}$	$2.159 \cdot 10^{-78}$
$\Pr[E] \approx 4/(3n)$	$1.338 \cdot 10^{-5}$	$1.151 \cdot 10^{-77}$
$\Pr[B]$	$\lesssim 10^{-9}$	negligible
$\Pr[C]$	$\approx 5.6 \cdot 10^{-4}$	$\approx 5.6 \cdot 10^{-4}$
$\Pr[D]$	≈ 0.249	≈ 0.249

Classes C and D keep their $\Theta(1)$ densities on any curve — and remain worthless, since neither yields a candidate. The two classes that do yield a key occur on secp256k1 with probability below 10^{-77} per signature. For scale: if every signature ever produced by every Bitcoin wallet amounted to 10^{10} , the expected number of class-A or class-E hits among them would be about 10^{-67} .

7 Audit of the shipped data

Every printed row of all four report files was re-parsed, re-classified with the current code, and re-verified on the curve. Results:

File	rows	valid sigs	class agrees	recov. attempts	verified
...232542 (test, 5k)	1003	1003	1003	0	0
...233004 (legacy, 5k)	1000	1000	1000	0	0
...003013 (test, 10^8)	2895	2895	2895	1566	896
...223812 (legacy, forged)	100	100	100	100	100

“Valid sigs” means all three of $s \equiv k^{-1}(z + rx)$, $(kG)_x \bmod n = r$ and $s_{zk} + s_{rxk} \equiv s$ hold. No row failed; no classification disagreed; no rejected candidate ever passed Lemma 2.4. Three observations follow, each of which contradicts something in the repository’s own documentation.

7.1 The class-E rows carry a detector fingerprint

All 670 class-E rows in `transaction_list_20260809003013.txt` satisfy $S \equiv 2 \pmod{4}$. Not one has $S \equiv 1$.

By Corollary 6.2 the current detector produces the two residues in equal numbers, so under the current code the probability of that observation is 2^{-670} . The file was produced by the earlier revision, which required S to be even and therefore discarded the entire $S \equiv 1$ family. Two independent confirmations:

- The exact census predicts 668.9 class-E signatures per 10^8 for the earlier detector and 1337.8 for the current one. The file reports 670.
- The constructed file `transaction_list_20260812223812.txt`, which is newer, splits its 50 class-E rows as 26 with $S \equiv 1$ and 24 with $S \equiv 2$ — the post-fix behaviour.

Consequence: the headline rate of $6.7 \cdot 10^{-6}$ quoted for class E in the repository’s README is the rate of the *old* detector. Re-running the same experiment with the current code would find about twice as many class-E signatures, at $1.34 \cdot 10^{-5}$.

7.2 The class-B row is missing from the counters

The same file reports Recovery Attempts: 1564 and Recovery Verified: 894. Those numbers decompose exactly as

$$1564 = \underbrace{224}_A + \underbrace{2 \times 670}_E, \quad 894 = 224 + 670,$$

leaving nothing for class B. Yet the file contains one class-B row, its `x_recovered` column is populated, and the re-run confirms that the class-B quadratic yields two candidates of which one verifies. The correct totals are therefore 1566 attempts and 896 verifications. The README compounds the discrepancy by reporting Case B Count: 0 and Total Observed Cases: 24 986 477, whereas the data file states 1 and 24 986 478, and by asserting under “Known limits” that class B has never been observed in a real run.

7.3 The HYP-001 reduction as documented is not an equivalence

See Theorem 4.8 and the remark following it. The documented form $\lfloor a/w \rfloor \in \{N, N-1\}$ is implied by (10) but does not imply it; the correct statement fixes which of the two values occurs, via the carry indicator δ .

8 What this is and is not

8.1 The honest reading

The formulas of Section 4 do not *find* s_{zk} . They name at most seven values out of n , and occasionally one of them is right. The observed rates — $1/(4n)$ for class A, $4/(3n)$ for class E — are exactly what random guessing of k^{-1} would produce. That is the entire result.

8.2 Specific limits

- **The class labels need secrets.** Classes A–D are defined through m_1 and m_2 , class E through $|s_{zk} - s_{rxk}|$. None of these can be evaluated from (z, r, s) . The only public object is the candidate set $\mathcal{C}$ and the on-curve test.
- **Classes C and D yield nothing.** m_1 is not a function of the public data; the level N is an input to `find_common_private_key.py`, not an output of it, and the module refuses `secp256k1` because one level window there is of size $\Theta(n)$.
- **Level-window intersection is not proof.** By (9) the bundled demo expects 1.016 coincidental survivors; measured over 150 random four-transaction bundles the unverified search returned about three impostors per run. Only Lemma 2.4 removes them.
- **The secp256k1 class-A and class-E rows are constructed.** Construction requires knowing x and k and choosing z ; the forged rows have `is_message_bound = False`. They demonstrate that the recovery formulas work at 256 bits, nothing more.
- **Test-mode z is not a message hash.** Test-mode triples satisfy every algebraic relation used here but would not verify against a message.
- **The classification is representative-dependent.** See Section 3.4. Class E is built on the choice $S \in \{s, s+n\}$, which is a statement about integer representatives, not about $\mathbb{Z}_n$.

- **Nothing transfers to deployed ECDSA.** On secp256k1 the events described here occur with probability below 10^{-77} per signature. Real-world ECDSA key recovery comes from nonce reuse, biased or partially-known nonces (the hidden-number problem and its lattice attacks), or faulty random number generators — none of which is what this taxonomy studies.

8.3 Reproduction

Claim	Command
Classifier and recovery	<code>python3 src/utis/generate_transactions.py</code>
Direct class construction	<code>... -tx-with-classes A B E</code>
Level-window intersection	<code>python3 src/utis/find_common_private_key.py</code>
Regression suite	<code>python3 tests/run_tests.py</code>

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